Semantic Change Detection Based on Supervised Contrastive Learning for High-Resolution Remote Sensing Imagery

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***Abstract*— Semantic change detection (SCD) for high-resolution remote sensing imagery involves simultaneously locating the changed regions and identifying the semantic change categories. Recently, a series of multitask Siamese networks have been proposed to model the SCD task by merging the binary change detection (BCD) results and the bitemporal land-cover classification (LCC) results. However, due to the large reflectance variability of the land cover in bitemporal high-resolution images, the land-cover feature clusters extracted by these methods are still considerably mixed, leading to the misidentification of change and their semantic changes types. In this article, to handle this problem, the SiamContrast method is proposed to learn temporally invariant discriminative land-cover features for SCD. As part of SiamContrast, a novel SCD contrastive loss (SCD-CL) is proposed to enhance the temporally invariant feature discrimination across bitemporal images. SCD-CL utilizes supervised contrastive learning and consists of two complementary components: mono-temporal contrastive loss (MCL) and cross-temporal contrastive loss (CCL). In particular, MCL contrasts the land-cover features within each temporal image, to enhance the mono-temporal feature discrimination. Meanwhile, CCL with a change-aware hard anchor sampling (CHAS) strategy contrasts the land-cover features across bitemporal images, to align the land cover features of the same category. To validate the effectiveness of SCD-CL, the SiamContrast method incorporates a difference feature pyramid (DFP) decoder, which leverages feature distance to model changes, allowing it to directly benefit from the discriminative features learned by SCD-CL. The comprehensive experimental results consistently confirm the effectiveness of the proposed SiamContrast in improving SCD performance, compared with the existing SCD methods. The code is available at https://github.com/wdczs/SiamContrast.**

***Index Terms*— Change detection, fully convolutional network (FCN), multitask learning, semantic change detection (SCD), semantic segmentation, Siamese network.**

I. Introduction

CHANGE detection (CD), which involves analyzing the changes of the objects of interest in an area through observing multitemporal high-resolution remote sensing images [1], [2], [3], [4], has been applied in many fields, such as land-cover monitoring [5], [6] and disaster emergency management [7], [8]. According to the extracted change information form, the CD technique can be further divided into binary change detection (BCD) and semantic change detection (SCD). Unlike the BCD task, which focuses only on whether a region is changed or not, the SCD task needs to identify not only the unchanged and changed areas but also the land-cover categories in the changed regions among the multitemporal images [9], [10].

There are four approaches to using deep learning for SCD tasks, including postclassification comparison, direct SCD, separate single-task learning, and multitask learning [9], [10]. Postclassification comparison method conducts land-cover classification (LCC) of bitemporal images, respectively, and then compares the classification results to obtain the results of SCD [10]. Peng et al. [11] proposed SCDNet, which even introduces change information represented by different features and spatial and channel attention to bitemporal LCC results, to enhance the PCC for SCD. However, due to the spectral differences between bitemporal images, the PCC can obtain an acceptable LCC accuracy, but obtains an extremely high recall and low precision when identifying change and no change [10], [12]. Direct SCD method treats each semantic change category as one "from-to" class, and performs fully end-to-end pixelwise classification to recognize the semantic change category [10], [13]. However, direct SCD method can face an extreme class imbalance of the "from-to" labels [9]. In these cases, the number of unchanged types greatly surpasses that of changed types, and the diversity among different change types also varies. This disparity can cause the model for "from-to" classification to fail to converge [12]. To overcome this challenge, both single-task and multitask learning approaches are proposed to first decompose the SCD task's "from-to" classification into a BCD task for identifying changed regions, and then apply a bitemporal LCC task to determine semantic change categories from the BCD results. The separate single-task learning method independently trains the BCD and LCC models. In contrast, the multitask learning method employs a multitask Siamese network that utilizes a task-shared Siamese encoder and two BCD and LCC decoders to perform both tasks, simultaneously. Multitask learning has the advantage of capturing common and useful information between two tasks, making the multitask Siamese

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network the most advanced method that has attracted many researchers.

In particular, the multitask Siamese network extracts bitemporal features from the bitemporal images using a Siamese encoder. The extracted features are then used by the BCD and LCC decoders to predict the BCD result and the bitemporal LCC results, respectively. Depending on the approach used to predict semantic change types, the multitask Siamese network method can be further categorized into two groups: independent prediction and time-related prediction. As illustrated in Fig. 1(a), the independent prediction method involves performing bitemporal LCC by utilizing the bitemporal features extracted from the Siamese encoder, respectively. Ding et al. [14] proposed SSCD-l, a simple and effective fully convolutional network (FCN)-based multitask Siamese network. Zheng et al. [12] introduced the ChangeMask framework, which employs an U-Net-based BCD decoder with a temporally symmetrical transformer to obtain temporally symmetric BCD features to locate change, and uses an U-Net-based LCC decoder to predict semantic change types. Shen et al. [17] proposed SFCCD, which employs multiscale feature fusion with global channel and spatial attention for the BCD prediction and LinkNet-based decoder [18] to obtain bitemporal LCC features for the semantic change prediction. Deng et al. [19] proposed MCDNet, which utilizes a gate to fuse two types of change features to obtain BCD results and use U-Net-based LCC decoder with channel attention to predict semantic change types. Li et al. [20] proposed DEFO-MTLSCD, which aims to enhance decoder feature interactions from the BCD to LCC subtasks, resulting in promising performance in SCD. To establish long-range dependencies in both spatial and temporal dimensions, Zhuang et al. [21] and Niu et al. [22] proposed the multitask learning networks with hybrid architectures based on CNN and transformer. In contrast, Chen et al. [23] introduced ChangeMamba, a model based on state space models that effectively captures global spatial contextual information. However, the lack of communication between the bitemporal LCC may limit the ability of these independent prediction methods to recognize semantic changes [14], [15].

As illustrated in Fig. 1(b), the time-related prediction method involves using a time relationship module to extract time-related features from the bitemporal features. These time-related features are then introduced to each temporal LCC, to predict semantic change type with temporal relationship. Ding et al. [14] proposed a Bi-SRNet, in which the changed regions are located by the FCN-based BCD decoder. Meanwhile, input bitemporal features, a cross-temporal semantic reasoning block based on attention mechanism acquires time-related features, which are then added to each temporal land-cover features, respectively, to predict the change type. Zhao et al. [15] proposed SSESN, and Cui and Jiang [24] proposed MTSCD-Net. Both models share a similar approach: BCD features are obtained using attention module, regarded as time-related features, and integrated into bitemporal LCC. More recently, Ding et al. [25] proposed SCanNet, a method to explore semantic information in settings with limited change samples, particularly focusing on cases where land-cover labels are available only in changed regions. In the SCanNet method, a SCanFormer is introduced to model deep and long-range temporal correlations for change and bitemporal features. Meanwhile, a semantic learning scheme is proposed to generate pseudo land-cover labels in unchanged regions, leveraging additional supervision to reduce discrepancies in bitemporal results.

However, as shown in Fig. 1, SSCD-l [14] and SSESN [15], which are representative independent prediction and time-related prediction methods, respectively, overlook the changed regions with small spectral differences, such as the change from bare land to grass in the center of the t_1 and t_2 images. Furthermore, even within the regions that have been identified as changed, bard land and grass are incorrectly recognized as water, road, and buildings, due to the presence of building shadows, resulting in erroneous semantic change predictions. These phenomena suggest that the state-of-the-art SCD methods still face the limitations when dealing with high-resolution remote sensing images that exhibit significant reflectance variability among various land covers. The feature visualization in Fig. 1 demonstrates that there is a significant overlap between different clusters of land-cover features due to reflectance variability. The presence of overlap poses limitations on the accuracy of binary change predictions and semantic change type recognition. Specifically, for the two types of multitask Siamese networks, two key factors can be identified.

1) In terms of BCD, both types of multitask Siamese networks utilize bitemporal land-cover features to derive BCD features. The confusion among these features can result in inaccurate binary change predictions.
2) In terms of change type prediction, independent prediction methods lacking temporal modeling directly face challenges in recognizing semantic change types due to the presence of overlapping features. Meanwhile, time-related prediction methods that employ time-related features may encounter cumulative errors during semantic change type recognition because overlapped features can obscure the semantics of the temporal features.

Hence, it is crucial to address the issue of feature cluster confusion caused by the substantial reflectance variability for SCD. This requires that land-cover features of the same type are similar in the mono-temporal images as well as in the different-temporal images. Meanwhile, land-cover features of different types are dissimilar in the two temporal images. Namely, temporally invariant discriminative land-cover features are needed for the SCD task in high-resolution remote sensing images.

Inspired by the recent advances in contrastive learning, supervised contrastive learning has the potential to enhance the feature discrimination for the SCD task. Supervised contrastive learning, which employs labels to define positive and negative samples and contrasts them in the embedding space [26], has been successful in learning pixel-level discriminative features for natural images in the semantic segmentation field [27]. The fundamental principle of contrasting samples is that an anchor feature is attracted by multiple positives while being repelled by multiple negatives [26]. In this context, supervised

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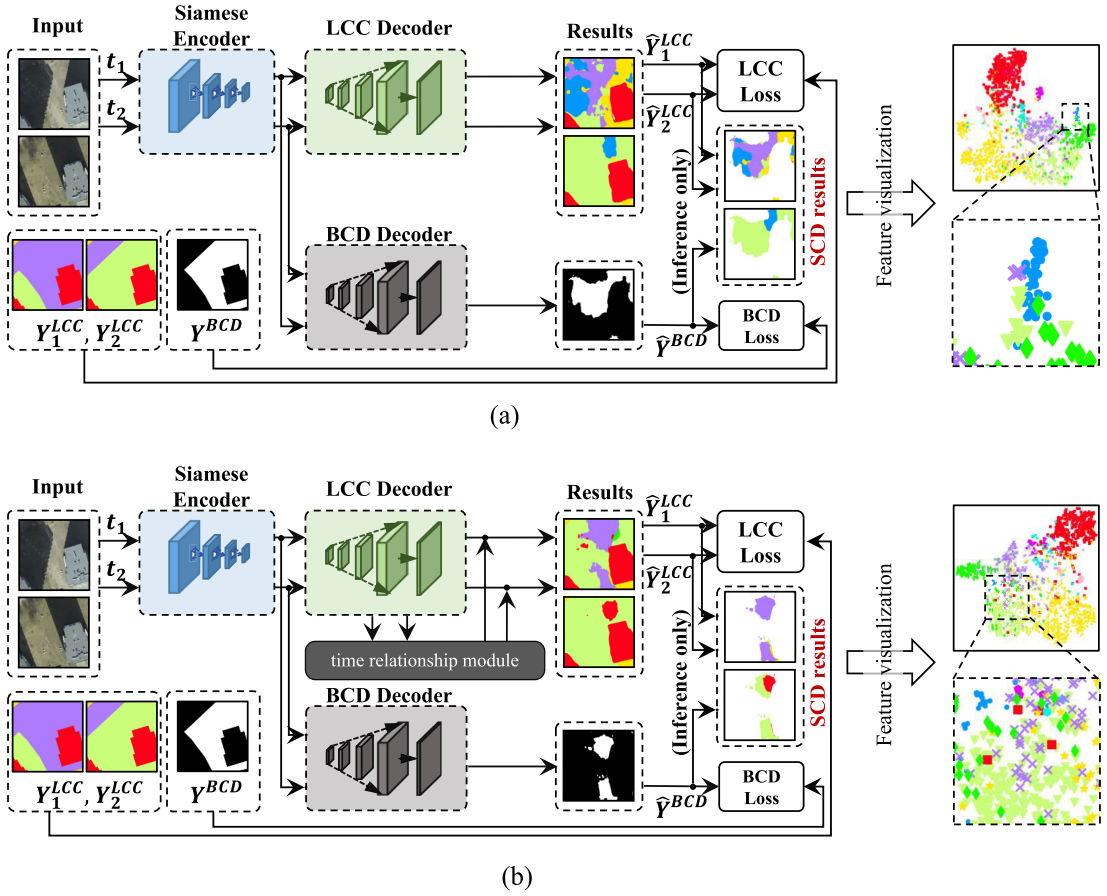


Fig. 1. Illustration of the two types of advanced multitask Siamese networks for SCD. Their representative methods, i.e., SSCD-l [14] and SSESN [15], were used to obtain SCD results and t-SNE visualizations on the Hi-UCD mini dataset [16], respectively. Here, the large spectral reflectance causes the confusion of bitemporal land-cover features, leading to the incorrect prediction of the changed locations and their semantic change types (■ water, ■ grass, ■ buildings, ■ greenhouse, ■ road, ■ bridge, ■ bare land, ■ woodland, and ■ others). (a) Independent prediction method. (b) Time-related prediction method.

contrastive learning offers two potential and direct approaches for improving the feature representation of bitemporal land covers. Following the principles of supervised contrastive learning used in semantic segmentation, the first approach involves contrasting features extracted from each temporal image, respectively, to reduce confusion among land-cover features within a mono-temporal phase. However, it fails to address the disparity in bitemporal land-cover feature distributions, consequently restricting its ability to identify land covers and semantic changes across bitemporal images. The second approach involves contrasting features extracted from two temporal images, where bitemporal features are given equal consideration to be contrasted. However, significant differences in bitemporal land-cover feature distributions can cause contrasting features across bitemporal images to contribute a significant proportion of contrastive loss when compared to within mono-temporal images. As a result, the feature discrimination in mono-temporal images is suppressed, which can limit the compactness of each land covers in bitemporal images.

In this article, to overcome the feature confusion problem caused by the large reflectance variability in bitemporal high-resolution remote sensing images, a novel SCD contrastive loss (SCD-CL) method based on supervised contrastive learning is proposed to learn temporally invariant discriminative land-cover features. To achieve this, the proposed SCD-CL method consists of two complementary components: mono-temporal contrastive loss (MCL) and cross-temporal contrastive loss (CCL), both of which are based on supervised contrastive learning. The MCL contrasts land-cover features within each temporal phase, to ensure the feature discrimination within the mono-temporal images. Simultaneously, the CCL contrasts among bitemporal features to align the land-cover features belonging to identical classes between the two time phases. In addition, a change-aware hard anchor sampling strategy is proposed for the CCL, to select hard samples from changed regions, to increase the discrepancy between features of different land-cover categories, and to enhance the semantic change recognition ability. Hence, the SCD-CL can effectively learn time-invariant discriminative features for land covers of bitemporal images.

To demonstrate the effectiveness of the proposed SCD-CL method, the SiamContrast method is proposed. This method utilizes a simple yet efficient architecture for SCD. To locate the changed regions, based on the feature pyramid network (FPN) [28], the SiamContrast method employs a difference feature pyramid (DFP) decoder, which utilizes feature distance to model change and no change. This approach makes the DFP decoder directly benefit from the discriminative features learned by the SCD-CL to obtain high-resolution BCD features. In line with the DFP decoder, the SiamContrast method incorporates a Siamese feature pyramid (SiamFP) decoder to

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acquire high-resolution bitemporal LCC features, to recognize the change regions's type. Therefore, unlike many recent advanced SCD methods that rely on intricate attention architectures for their BCD and LCC, the SiamContrast method can effectively use discriminative feature to obtain promising SCD performance without a high computational complexity.

The main contributions of this article are as follows.

1) To address the deep feature confusion problem caused by the complex bitemporal high-resolution imagery, the SiamContrast method is introduced, integrating the SCD-CL from supervised contrastive learning with a DFP decoder. This approach aims to learn temporally invariant discriminative land-cover features and effectively model change for the SCD task.
2) To enhance the temporally invariant feature discrimination, SCD-CL is proposed, incorporating complementary mono-temporal (MCL) and cross-temporal (CCL) contrastive losses, coupled with a change-aware hard anchor sampling strategy, to minimize intraclass distance and maximize interclass distance for bitemporal land-cover features.

The proposed SiamContrast method was conducted on two public SCD datasets. The quantitative and visualization experimental results show that the proposed approach can achieve a better SCD performance than the advanced comparison methods.

The rest of this article is organized as follows. Section II briefly revisits the recent deep learning-based BCD models, supervised contrastive learning methods, and contrastive learning-based CD methods Section III describes the proposed SiamContrast method in detail. The quantitative and visualization experimental results of the proposed SiamContrast obtained on the two SCD datasets are provided in Section IV. In Section V, an ablation study is provided. Finally, our conclusions are given in Section VII.

II. Related Work

A. BCD Models Based on Deep Learning

Due to the complex characteristics of the land cover in bitemporal high-resolution remote sensing images and the ability of deep learning to extract robust features, researchers are now focusing on deep learning-based BCD techniques [2], [29].

According to the differences in computing units on the network, BCD methods based on deep learning can be divided into two categories: those based on convolutional neural networks (CNNs) [29], [30], [31], [32], [33], [34], [35], [36] and those based on transformers [3], [37], [38], [39]. CNNs are good at extracting detailed information, while transformers are good at capturing global image information [38], [39]. Most BCD methods, whether based on CNN or transformer, have a main structure of Siamese network consisting of a weight-shared Siamese encoder, which extracts bitemporal features from both images, and decoder, which uses the bitemporal features to predict the change. In this context, whether it is an independent BCD task or a BCD subtask in SCD, it is crucial for accurate BCD to utilize bitemporal features to model changes in the decoder.

There are two approaches to deriving BCD features from bitemporal features: hand-crafted measures [29], [30], [31] and automatic learning [12], [15], [19], [33], [34], [38]. The hand-crafted measure involves calculating feature differences between bitemporal features using methods such as L_1 or L_2 distances [29], [30], [31]. In these methods, greater differences in features indicate a higher likelihood of changes occurring in the land covers. While this method provides interpretability, it may limit the effectiveness of representing changes due to its reliance on hand-crafted measures [12]. To enhance the capability of BCD features, advanced methods have been proposed for automatic learning. Many of these methods utilize temporally concatenated bitemporal features [14], [29], with some even incorporating attention mechanisms to improve their performance [3], [17], [29], [32], [35], [36], [37], [39]. However, this representation approach is temporally asymmetric, leading to potential overfitting of the temporal order of concatenated bitemporal features by the models [12], [19]. To derive BCD features in a temporally symmetric manner, some researchers have proposed applying attention mechanisms to the calculation of hand-crafted measure methods [15], [19], [33], [34], [38]. Additionally, Zheng et al. [12] introduced learnable temporal-symmetric operators, specifically the temporal-symmetric transformer. While automatic learning BCD features offer flexible and powerful representational capabilities, it may overlook the inherent causal relationship between bitemporal features and BCD features, potentially leading to less interpretable results [40].

In the SiamContrast method, the proposed SCD-CL method has the ability to learn temporally invariant discriminative features, which considerably matches the approach of the absolute subtraction operator to model change. This operator is a representative hand-crafted feature difference operator. By employing this operator, the explicit relationship between bitemporal land cover and BCD features can be accurately described with feature distances, resulting in the acquisition of interpretable and powerful representations of BCD features.

B. Supervised Contrastive Learning

Contrastive learning, the core idea of which is "learn to compare" [41], is an effective representation learning method for neural networks. To perform the core idea, recent contrastive learning methods map the features to projection space, where an anchor is pulled by the positives and pushed away by many negatives [26], [42], [43].

In the unsupervised contrastive learning methods, an image is augmented twice to obtain its multiple views as a positive pair, while the other images are used as negative samples [42], [44], the noise of which may lead to overfitting. To overcome this problem, supervised contrastive learning uses the semantic labels of the samples to define positives and negatives [26]. In this context, multiple positives and negatives are used for an anchor to contrast in the projection space, which accounts for the effectiveness of supervised contrastive learning in enhancing the feature discrimination [26], [27]. Supervised contrastive learning can also be regarded as generalized triplet loss [45].

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Many researchers [26], [46], [47] have used supervised contrastive learning for pretraining, but this paradigm requires additional pretraining time before training a model for the downstream task. To overcome this, other researchers have proposed to apply supervised contrastive learning as a regularization term to improve the feature discrimination in an end-to-end manner [27], [48], [49]. To handle the imbalanced fineness and appearances for fine-grained ship classification, Chen et al. [49] proposed a push-and-pull network, in which the neural network is trained by the sum of the supervised contrastive loss and the cross-entropy classification loss to learn discriminative features. Huang et al. [48] combined supervised contrastive learning and hash quantization to enhance the feature discrimination for remote sensing image retrieval. Wang et al. [27] proposed pixel-to-pixel contrastive learning, which regards supervised contrastive learning as a regularization term for training an FCN [50] for semantic segmentation, to improve the pixel-level feature discrimination.

C. Change Detection Based on Contrastive Learning

One of the major challenges in developing deep learning-based CD models is the high expense of acquiring labeled bitemporal images for training. To address this challenge, researchers have adopted contrastive learning techniques that pretrain the encoder to acquire outstanding representation ability [4], [46], [51], [52], [53], [54], [55], [56], [57]. Consequently, contrastive learning-based methods can yield robust initialization parameters for the encoder, which subsequently enhances the performance of CD models.

Contrastive learning-based pretraining methods for CD can be categorized into two types based on the use of labels: unsupervised and supervised [51], [52], [53], [54], [55], [56]. In unsupervised contrastive learning pretraining methods, it is assumed that the same geographical location in multitemporal images remains unchanged, and the unchanged land covers are considered positive samples, while those from different locations are negative samples. By pretraining on large quantities of unlabeled multitemporal remote sensing data, strong encoder parameters for initialization can be acquired [51], [55], [56]. Alternatively, supervised contrastive learning pretraining methods utilize labels to construct positive and negative samples, allowing the encoder to learn representation ability directly from labeled data [4], [46].

In short, the existing CD methods based on contrastive learning aim to improve the representation ability of encoder initialization parameters and further require end-to-end training for SCD task. In contrast, this article introduces SCD-CL, which employs the concept of "contrast" as an auxiliary loss to learn discriminative features between bitemporal images while end-to-end training SCD models.

III. Proposed SiamContrast

SiamContrast is proposed to learn temporally invariant discriminative land-cover features for bitemporal images. The proposed SiamContrast utilizes supervised contrastive learning, which can aggregate the positives and separate the negatives of the bitemporal land-cover features effectively, to enhance the feature discrimination of bitemporal images and improve the SCD ability. Meanwhile, the proposed DFP encoder and SiamFP decoder ensure the high computational efficiency of the whole SiamContrast, due to the efficiency of the FPN.

With input bitemporal images, SiamContrast first performs BCD and bitemporal LCC by the DFP and SiamFP decoder. The model is then optimized by the joint loss of BCD cross-entropy loss, LCC cross-entropy loss, and SCD-CL in the training phase. In the testing phase, the BCD result is used as a mask to determine where the changes are, and the bitemporal LCC results are combined to determine the semantic change types of the changed regions that the BCD result has identified.

A. Overview of SiamContrast

As shown in Fig. 2, the proposed SiamContrast is defined by a set of learnable parameters θ and consists of three learnable components: the Siamese encoder E_θ, the DFP decoder D_θ^{DFP}, and the SiamFP decoder D_θ^{SFP}.

As shown in (1), for given bitemporal remote sensing images t_1 and t_2, they are first fed into the parameter-shared Siamese encoder E_θ to obtain the spatial feature sets F_1 for the t_1 image and F_2 for the t_2 image

$$F_1 = E_\theta(t_1),\ F_2 = E_\theta(t_2) \tag{1}$$

where the feature sets $F_1 = \{f_1^{(1)}, f_1^{(2)}, f_1^{(3)}, f_1^{(4)}\}$ and $F_2 = \{f_2^{(1)}, f_2^{(2)}, f_2^{(3)}, f_2^{(4)}\}$ contain feature maps $f^{(*)}$ of different spatial resolutions $*$. Each feature map is chosen from the convolution layer following the downsampling layer in the Siamese encoder. With an input image size of 512, the sizes of the aforementioned feature maps are 256, 128, 64, and 32, respectively.

Based on the FPN [28], the DFP decoder is proposed to leverage the characteristics of the features learned by SCD-CL to model change for change map predictions. As shown in (2), F_1 and F_2 are fed simultaneously into the DFP decoder D_θ^{DFP} to obtain the BCD result $\hat{Y}^{\mathrm{BCD}}$

$$\hat{Y}^{\mathrm{BCD}} = D_\theta^{\mathrm{DFP}}(F_1, F_2). \tag{2}$$

To efficiently obtain the robust high-resolution land-cover feature for each temporal image, the weight-sharing SiamFP decoder based on an FPN [28] is used to fuse the multiresolution features of the single-temporal image. As shown in (3), bitemporal land-cover maps $\hat{Y}_1^{\mathrm{LCC}}$ and $\hat{Y}_2^{\mathrm{LCC}}$ are respectively predicted by their multiresolution feature sets F_1 and F_2 by the SiamFP decoder D_θ^{SFP}

$$\hat{Y}_1^{\mathrm{LCC}} = D_\theta^{\mathrm{SFP}}(F_1),\ \hat{Y}_2^{\mathrm{LCC}} = D_\theta^{\mathrm{SFP}}(F_2). \tag{3}$$

The proposed SCD-CL is used together with the BCD loss and LCC loss in the training phase. In this sense, the loss and the optimization of SiamContrast can be described as follows:

$$L = L_{\mathrm{BCD}} + L_{\mathrm{LCC}} + \lambda L_S \tag{4}$$

$$\theta^{(k+1)} = \theta^{(k)} - \eta \frac{\partial L}{\partial \theta^{(k)}} \tag{5}$$

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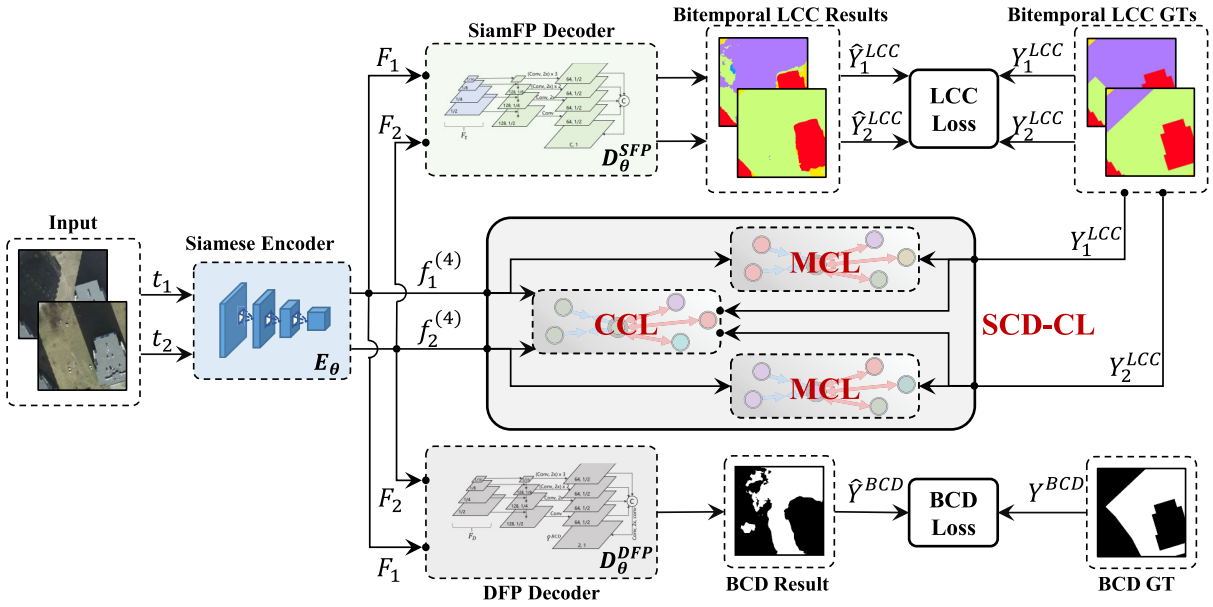


Fig. 2. Overview of the proposed SiamContrast. For the input bitemporal images for SiamContrast, the task-shared Siamese encoder respectively extracts bitemporal feature map sets F_1 and F_2, which are then used to extract the BCD features and output the BCD result $\hat{Y}^{\mathrm{BCD}}$ by the DFP decoder. Meanwhile, with inputs F_1 and F_2, the SiamFP decoder outputs bitemporal LCC predictions $\hat{Y}_1^{\mathrm{LCC}}$ and $\hat{Y}_2^{\mathrm{LCC}}$, so as to recognize semantic changes. To optimize the whole model, the lowest-resolution bitemporal feature maps $f_1^{(4)}$ and $f_2^{(4)}$ extracted by the Siamese encoder are mapped to projections, which are used to compute SCD-CL with labels. At the same time, the BCD and LCC predictions and their labels are used to compute the BCD loss and LCC loss.

where λ denotes the balance coefficient between the proposed SCD-CL, and k denotes the k-th iteration. The cross-entropy losses for the BCD and LCC tasks are computed as follows:

$$L_{\mathrm{BCD}} = CE\left(\hat{Y}^{\mathrm{BCD}}, Y^{\mathrm{BCD}}\right)$$
$$L_{\mathrm{LCC}} = \left(CE\left(\hat{Y}_1^{\mathrm{LCC}}, Y_1^{\mathrm{LCC}}\right) + CE\left(\hat{Y}_2^{\mathrm{LCC}}, Y_2^{\mathrm{LCC}}\right)\right)/2 \quad (6)$$

where CE is the cross-entropy loss, and Y^{BCD}, Y_1^{LCC}, and Y_2^{LCC} are the ground truths of the BCD and LCC tasks. In the test phase, the sth ($s = 1$ or 2) temporal SCD result $\hat{Y}_s^{\mathrm{SCD}}$ is obtained by the result merging strategy as follows:

$$\hat{Y}_s^{SCD} = \begin{cases} \text{non-change}, & \text{if } \hat{Y}^{\mathrm{BCD}} = \text{non-change} \\ \hat{Y}_s^{\mathrm{LCC}}, & \text{if } \hat{Y}^{\mathrm{BCD}} = \text{change}. \end{cases} \quad (7)$$

B. DFP Decoder

The DFP decoder is utilized to identify the locations where changes have occurred based on the characteristics of the features acquired through the SCD-CL method. The primary objective of the proposed SCD-CL is to amplify the interclass distance between different land cover types and diminish the intraclass distance within the same land cover type across bitemporal features. This objective aligns with the approach of the feature difference operator in modeling change. More specifically, the feature difference operator signifies significant feature distances to represent changes between land covers of different types, while indicating small feature distances to represent no change within the same land cover type. Consequently, the proposed SCD-CL can directly and effectively enhance the ability of the feature difference operator in representing both change and no change. To predict the binary change map accurately, an FPN architecture, which is an effective decoder used to obtain a robust high-resolution feature representation in the semantic segmentation field [28], is used to fuse the multiresolution BCD features.

In this context, the DFP decoder can be divided into two main steps: 1) obtaining multiresolution BCD features and 2) obtaining high-resolution BCD feature and predicting.

1) Obtaining Multiresolution BCD Features: The absolute subtraction operator is a representative feature difference operator. In this context, the BCD features are modeled by the absolute subtraction operator between bitemporal spatial features

$$f_D^{(*)} = |f_1^{(*)} - f_2^{(*)}| \quad (8)$$

where $f_1^{(*)}$ and $f_2^{(*)}$ are feature maps of the same spatial size. According to (8), multiresolution BCD feature set $F_D = \{f_D^{(1)}, f_D^{(2)}, f_D^{(3)}, f_D^{(4)}\}$ can be obtained from the bitemporal feature sets F_1 and F_2.

2) Obtaining High-Resolution BCD Feature and Predicting: As shown in Fig. 3, the multiresolution BCD features of F_D are first upsampled gradually, to obtain the DFP. Each feature map of DFP is then resized to the same size by several upsampling units, each of which includes 3×3 convolution, group normalization [58], rectified linear unit (ReLU), and bilinear upsampling operations. These 64-D resized high-resolution BCD features are then concatenated and upsampled to the spatial size of the input bitemporal images, to predict the change map $\hat{Y}^{\mathrm{BCD}}$. The pseudo code for the DFP decoder is presented in Algorithm 1.

C. SiamFP Decoder

For convenience, the architecture of the LCC decoder in the SiamContrast also uses FPN-based decoder, i.e., SiamFP decoder. The differences between the DFP and SiamFP

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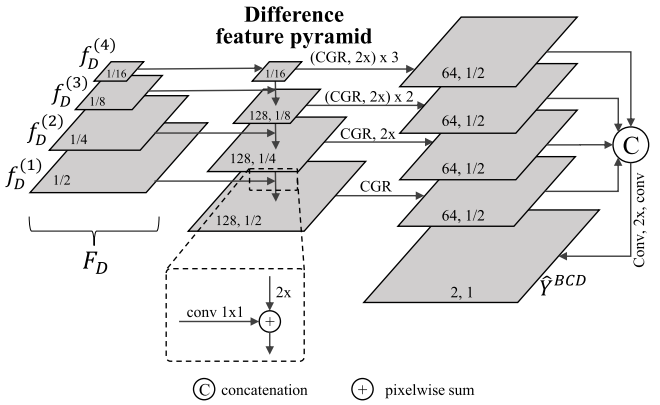


Fig. 3. Illustration of the proposed DFP decoder. With the input multiresolution BCD feature set F_D, the FPN architecture upsamples features to recover their spatial resolution. These features are then upsampled to the same size and concatenated to obtain the high-resolution BCD features and predict the change map $\hat{Y}^{BCD}$. "CGR" denotes "convolution-group norm-ReLu", and "2×" denotes 2× bilinear upsampling.

Algorithm 1 Processing Procedure of the DFP Decoder

Input: multiscale bitemporal feature sets $F_1 = \{f_1^{(k)}\}_{k=1}^4$ and $F_2 = \{f_2^{(k)}\}_{k=1}^4$
for each $k \in [1, 4]$ **do**
  $f_D^{(k)} = |f_1^{(k)} - f_2^{(k)}|$
end for
$F_D = \{f_D^{(k)}\}_{k=1}^4$
$\hat{Y}^{BCD}$ =FPN(F_D)
Return: BCD prediction $\hat{Y}^{BCD}$

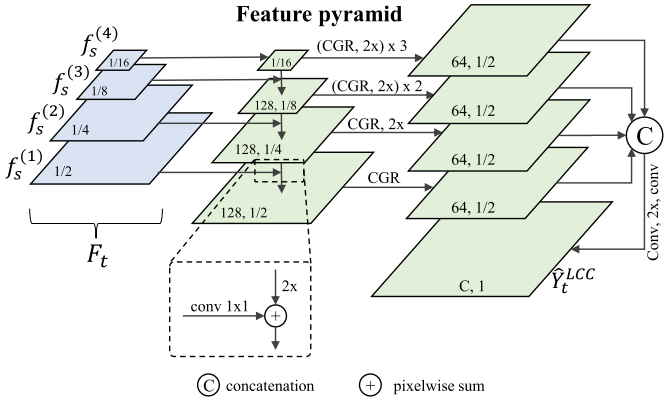


Fig. 4. Illustration of the SiamFP decoder. With the input sth temporal multiresolution feature set F_s, the FPN architecture upsamples the features to recover the spatial resolution of the features, which are then upsampled to the same size and concatenated to predict the sth temporal LCC map $\hat{Y}_s^{LCC}$. "CGR" denotes "convolution-group norm-ReLu", and "2×" denotes 2× bilinear upsampling.

decoders lie in the input features and output channels. In this context, the weight-sharing SiamFP decoder obtains high-resolution bitemporal land-cover feature maps to predict land-cover maps from the bitemporal input feature sets F_1 and F_2 in an efficient manner.

As shown in Fig. 4, the sth temporal multiresolution features of F_s are first upsampled gradually, to obtain its feature pyramid. Using the upsampling unit composed of 3×3 convolution, group normalization [58], ReLU, and bilinear upsampling operations, each feature map of the feature pyramid is then resized to the same size, with their dimension reducing to 64. Finally, these features are concatenated and upsampled to the spatial size of the input bitemporal images, to predict the LCC map $\hat{Y}_s^{LCC}$ of C classes for the sth temporal image.

D. SCD Contrastive Loss

1) Overview of SCD-CL: SCD-CL is proposed to learn temporally invariant discriminative features for bitemporal images. As shown in Fig. 5, SCD-CL includes two complementary parts, i.e., MCL and CCL. In particular, inspired by the literature [27], the MCL performs contrastive learning using features from mono-temporal images, i.e., t_1 and t_2 images, respectively, to boost the feature discrimination. CCL contrasts between features from both temporal images to align the single-temporal land-cover feature with that of the other, to ensure the correctness of the semantic change prediction. The CCL is weighted and added to the MCL in the proposed SCD-CL method

$$L_S = (L_{MC} + \mu L_{CC})/2 \tag{9}$$

where μ denotes the hyperparameter to control the ratio of the CCL.

In line with many contrastive learning methods [27], [42], [43], [44], both the MCL and CCL are performed on the lowest-resolution feature maps output by the top of the encoder. We take the first temporal image as an example. The lowest-resolution feature map $f_1^{(4)}$ in F_1 of the first temporal image is of size $H \times W \times C$. To guarantee the quality of the features learned by contrastive learning, the features are usually mapped to projection space, where the contrastive learning is performed [27], [42], [43]. Here, $f_1^{(4)}$ is mapped to the projection map by a projector of "conv-batchnorm-ReLu-conv" architecture [27], [43]. The projection map is of size $H \times W \times 128$, which includes $H \times W$ of 128-D projection vectors. In this context, $\boldsymbol{v}_{1,i}$ denotes the ith projection vector for the t_1 images in the mini-batch. Similarly, we have $\boldsymbol{v}_{2,i}$, which denotes the ith projection vector for the t_2 images in the mini-batch. Given the land-cover labels of bitemporal images, each of their projections serves as an anchor and a positive sample for projections within its own class, while concurrently acting as a negative sample for projections belonging to different land-cover classes. For convenience, Fig. 5 only displays one anchor being pulled by its positive samples and pushed away from samples of other classes both in the MCL and CCL.

2) MCL of SCD-CL: As shown in Fig. 5, in the MCL, the land-cover projections of each temporal image are contrasted with their respective temporal projections, which means that the positives and negatives for the contrastive learning come from only single-temporal images, to learn discriminative features for the mono-temporal images. In this context, the MCL is made of two independent losses for the bitemporal images

$$L_{MC} = (L_{MC1} + L_{MC2})/2. \tag{10}$$

We take L_{MC1} for the first temporal image as an example. To enhance the discriminative power of land-cover features in the MCL, a segmentation-aware hard anchor sampling (SHAS) strategy is implemented, as suggested by Wang et al. [27].

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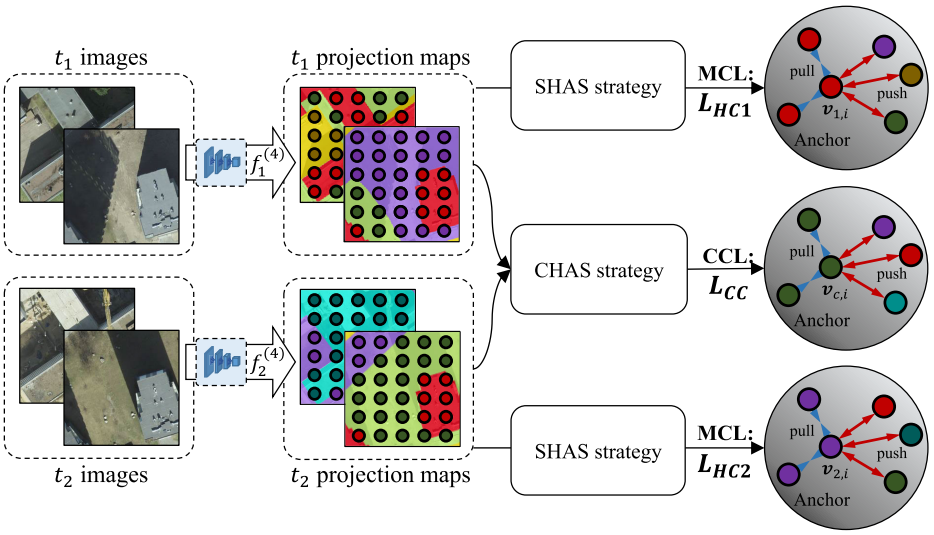


Fig. 5. Illustration of the proposed SCD-CL. Land-cover features are extracted from bitemporal images t_1 and t_2 by the Siamese encoder E_θ, and then mapped to their projection maps. Here, the land-cover projections of the mono-temporal images are first selected by the existing SHAS strategy, and are then contrasted to improve the mono-temporal feature discrimination in L_{MC1} and L_{MC2}. Meanwhile, the land-cover projections of the bitemporal images are first selected by the proposed change-aware hard anchor sampling (CHAS) strategy, and are then contrasted to align the features of the land covers in L_{CC}. The circles of different colors denote projections of different land covers.

In this approach, the focus is placed on ensuring that the hard anchors are prioritized, as they contribute more gradients compared to the other easy anchors. The emphasis on hard anchors is achieved by controlling their proportion among all the anchors in the MCL. To achieve this, a subset of projections from the t_1 image is utilized. Specifically, for the MCL, each land-cover category in each t_1 image has a selection of M projections sampled. Of these M projections, half $(M/2)$ are hard anchors, while the remaining half are easy anchors. If the number of hard or easy anchors falls below $M/2$, additional easy or hard anchors are sampled to ensure a total of M anchors. Within this framework, hard anchors are defined as the projections corresponding to the inaccurately predicted pixels in the LCC of the t_1 images. Here, M is a predefined hyperparameter. In this context, within a mini-batch of t_1 images, the MCL calculation involves a total of A_1 projections. The value of A_1 is computed as follows:

$$A_1 = \sum_{b=1}^{B} C_1(b) \cdot M \tag{11}$$

where B denotes the batch size, and $C_1(b)$ denotes the number of land-cover classes for the bth image of B of t_1 images. Hence, based on InfoNCE loss [42], [44], the MCL for the ith projection anchor in the t_1 images can be formulated as follows:

$$(L_{MC1})_i = \frac{-1}{|\Phi_1(i)|} \sum_{\varphi}^{\Phi_1(i)} \log \frac{\exp\left(\boldsymbol{v}_{1,i} \cdot \boldsymbol{v}_{1,\varphi}/\tau\right)}{\sum_{j,j\neq i}^{A_1} \exp\left(\boldsymbol{v}_{1,i} \cdot \boldsymbol{v}_{1,j}/\tau\right)} \tag{12}$$

where $\tau > 0$ is a predefined temperature hyperparameter set to 0.1, $\Phi_1(i) \equiv \{y_{1,\varphi} = y_{1,i}, \varphi \neq i\}$ is the index set of the selected $|\Phi_1(i)|$ positives of the ith projection for its land cover in the t_1 images of the mini-batch, and $y_{1,i}$ denotes the land-cover label of $\boldsymbol{v}_{1,i}$. Correspondingly, the MCL for the t_1 images of the mini-batch can be represented as follows:

$$L_{MC1} = \frac{1}{A_1} \sum_{i}^{A_1} (L_{MC1})_i. \tag{13}$$

Similarly, the MCL for the ith projection in the t_2 images of the mini-batch can be formulated as follows:

$$(L_{MC2})_i = \frac{-1}{|\Phi_2(i)|} \sum_{\varphi}^{\Phi_2(i)} \log \frac{\exp\left(\boldsymbol{v}_{2,i} \cdot \boldsymbol{v}_{2,\varphi}/\tau\right)}{\sum_{j,j\neq i}^{A_2} \exp\left(\boldsymbol{v}_{2,i} \cdot \boldsymbol{v}_{2,j}/\tau\right)}$$

$$L_{MC2} = \frac{1}{A_2} \sum_{i}^{A_2} (L_{MC2})_i \tag{14}$$

where $\Phi_2(i) \equiv \{y_{2,\varphi} = y_{2,i}, \varphi \neq i\}$ is the index set of the selected $|\Phi_2(i)|$ positives of the ith projection for its land cover in the t_2 images of the mini-batch, and $y_{2,i}$ denotes the land-cover label for anchor $\boldsymbol{v}_{2,i}$. A_2 denotes the number of selected projections in the t_2 images of the mini-batch.

In this context, minimizing (13) and (14) is equivalent to increasing the cosine similarity between each anchor $\boldsymbol{v}_{s,i}$ and its positive sample $\boldsymbol{v}_{s,\varphi}$, and reducing the cosine similarity between anchor $\boldsymbol{v}_{s,i}$ and its negative sample $\boldsymbol{v}_{s,j}$, where s is time phase 1 or 2. By doing this, the intraclass distance is reduced and the interclass distance is increased within mono-temporal images.

3) CCL of SCD-CL With CHAS Strategy: As shown in Fig. 5, in the CCL, the land-cover projections are contrasted across bitemporal images, to align the features of the same class in the different temporal images. Here, to enhance the recognition capability of change types, the change-aware hard anchor sampling (CHAS) strategy is proposed to focus the CCL only on hard anchors in the *changed* regions and enhancing the bitemporal feature representation. In principle, as in SHAS strategy [27], considering any misclassified sample in any region as a hard anchor in bitemporal imagery can enhance the overall feature representation and improve

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change recognition capabilities. However, in advanced multitask Siamese networks, the determination of change or no change relies on BCD, while the recognition of change type is carried out by LCC. Within the unchanged regions, there is a high likelihood that numerous pixels misclassified by LCC can be accurately identified as unchanged by BCD. Thus, treating these pixels in the unchanged regions as hard anchors prioritizes the minimization of intraclass distances across bitemporal images. As a result, this particular approach can be ineffective for BCD. Most importantly, it can also diminish the potential increase in interclass distance, ultimately suppressing the overall performance of change type recognition.

In contrast, the proposed CHAS strategy involves treating pixels that are not correctly recognized by the LCC decoder in the changed area as hard anchors, the ratio of which are then controlled to emphasize their effect in CCL. For the selected hard anchors in the changed area, they mainly exist to reduce the distance between samples with the same class features within single time phase and increase the distance between samples with different classes across the two time phases. Thus, this effectively enhances the differentiation between features of various land-cover categories and promotes the performance of identifying change types.

To ensure that a significant proportion of the hard anchors come from the changed regions, a subset of projections is utilized from the bitemporal images. M_c projections are selected from each land-cover category in each temporal image. Among these projections, half ($M_c/2$) are hard anchors from the changed regions, while the remaining half are easy anchors selected from both changed and unchanged regions. If the number of hard or easy anchors falls below $M_c/2$, additional easy or hard anchors are sampled to maintain a total of M_c anchors. In this context, hard anchors refer to the projections that correspond to inaccurately predicted pixels in the LCC of the changed regions in the bitemporal images. The value of M_c is a predefined hyperparameter. Therefore, within a mini-batch of bitemporal images, the calculation of the CCL involves a total of A projections. The value of A is computed as follows:

$$A = \sum_{b=1}^{B} [C_1(b) + C_2(b)] \cdot M_c \tag{15}$$

where B denotes the batch size, and $C_1(b)$ and $C_2(b)$ denote the number of land-cover classes for the bth image of B of t_1 and t_2 images, respectively. We let $\boldsymbol{v}_{c,i}$ denote the ith projection vector from the projection maps of the bitemporal images. In this context, based on InfoNCE loss, the CCL for the ith projection of the bitemporal images of the mini-batch can be formulated as follows:

$$(L_{CC})_i = \frac{-1}{|\Phi_c(i)|} \sum_{\varphi}^{\Phi_c(i)} \log \frac{\exp\left(\boldsymbol{v}_{c,i} \cdot \boldsymbol{v}_{c,\varphi}/\tau\right)}{\sum_{j,j\neq i}^{A} \exp\left(\boldsymbol{v}_{c,i} \cdot \boldsymbol{v}_{c,j}/\tau\right)} \tag{16}$$

$$L_{CC} = \frac{1}{A} \sum_{i}^{A} (L_{CC})_i \tag{17}$$

where $y_{c,i}$ denotes the land-cover label of anchor $\boldsymbol{v}_{c,i}$, and $\Phi_c(i) \equiv \{y_{c,\varphi} = y_{c,i}, \varphi \neq i\}$ is the index set of the selected $|\Phi_c(i)|$ positives of the ith projection for its land cover in bitemporal images of the mini-batch, and $y_{c,i}$ denotes the land-cover label for anchor $\boldsymbol{v}_{c,i}$.

Similarly, minimizing (17) is equivalent to increasing the cosine similarity between each anchor $\boldsymbol{v}_{c,i}$ and its positive sample $\boldsymbol{c}_{c,\varphi}$, and reducing the cosine similarity between anchor $\boldsymbol{v}_{c,i}$ and its negative sample $\boldsymbol{c}_{c,j}$. By doing this, the intraclass distance is reduced and the interclass distance is increased across bitemporal images. The pseudo code for SCD-CL is presented in Algorithm 2.

Algorithm 2 Processing Procedure of SCD-CL

Input: 128-dimensional projection sets $\{v_{s,i}(b)\}_{i=1}^{H\times W}$, number of land-cover classes $Cs(b)$, LCC predictions $\hat{Y}_s^{LCC}(b)$, LCC ground truths $Y_s^{LCC}(b)$, and BCD ground truths $Y^{BCD}(b)$ for each s-th temporal image in the b-th bitemporal pair within batch size B
for each $b \in [1, B]$ **do**
  /*for the b-th pair*/
  for each $s \in [1, 2]$ **do**
    /*for the s-th time phase*/
    for each $k \in [1, C_1(b)]$ **do**
      /*for the k-th land-cover class*/
      /* SHAS strategy in the MCL*/
      From the set $\{v_{s,i}(b)\}_{i=1}^{H\times W}$, select $M/2$ projections at locations where $\hat{Y}_s^{LCC}(b) \neq Y_1^{LCC}(b)$, with the other $M/2$ chosen randomly
      /* CHAS strategy in the CCL*/
      From the set $\{v_{s,i}(b)\}_{i=1}^{H\times W}$, select $M_c/2$ projections at locations where both $\hat{Y}_1^{LCC}(b) \neq Y_1^{LCC}(b)$ and $\hat{Y}_2^{LCC}(b) \neq Y_2^{LCC}(b)$ occur in the changed regions of $Y^{BCD}(b)$, and randomly choose the remaining $M_c/2$ projections
    end for
  end for
end for
A_1 and A_2 projections, selected through the SHAS strategy for bitemporal images, are utilized to compute L_{MC1} and L_{MC2} of MCL, respectively
A projections chosen via the CHAS strategy for bitemporal images are employed to calculate L_{CC} of CCL
$L_S = (L_{MC1} + L_{MC2})/2 + \lambda + L_{CC}$
return: SCD-CL L_S

IV. Experiments

Three public datasets—the WHU building dataset [59], Hi-UCD mini dataset [16], and Hi-UCD dataset [60]—were used to validate the proposed method.

A. Experimental Settings

1) Benchmark Methods and Implementation Details: The comparison methods were recent advanced SCD methods, i.e., HRSCD.str4 [10], SCDNet [11], ChangeMask [12], SSCD-l [14], Bi-SRNet [14], SSESN [15], SFCCD [17], MCDNet [19], and SCanNet [25]. To compare these SCD methods fairly, following [11] and [14], the ResNet-34 backbone [61] was used for all the benchmark methods and the

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proposed method. The code implementation of the Siamese encoder and decoders references the SMP code base [62].

All the models were trained with the stochastic gradient descent (SGD) algorithm with the same initial learning rate of 0.01 and weight decay of 4E-4 on the three SCD datasets. For the WHU building and Hi-UCD mini dataset, mini-batch size was 4 (image pairs), the total number of training iterations was 20 000, and the learning rate was decreased by 10 at 15 000 and 19 000 iterations. For the large-scale Hi-UCD dataset, mini-batch size was 8, the total number of training iterations was 80 000, and the learning rate was decreased by 10 at 50 000 and 70 000 iterations. The checkpoints that achieved the best separated Kappa (SeK) coefficients on the validation set were further tested on the test set to obtain the final evaluation results. All the experiments were implemented in PyTorch 1.8 [63] on an NVIDIA Titan RTX graphics processing unit (GPU).

2) Evaluation Metrics:

1) *i) BCD Metrics:* In this article, following recent studies [9], [12], we adopt the intersection over union (IoU) of the change and the F1-score to evaluate the BCD result in the SCD results as follows:

$$\mathrm{IoU}_{ch} = \mathrm{TP}/(\mathrm{TP}+\mathrm{FN}+\mathrm{FP}) \tag{18}$$

$$\begin{cases} \mathrm{F1} = (2\cdot \mathrm{Pr}\cdot \mathrm{Re})/(\mathrm{Pr}+\mathrm{Re}) \\ \mathrm{Pr} = \mathrm{TP}/(\mathrm{TP}+\mathrm{FP}) \\ \mathrm{Re} = \mathrm{TP}/(\mathrm{TP}+\mathrm{FN}) \end{cases} \tag{19}$$

where TP, FN, and FP denote the numbers of true positive, false negative, and false positive pixel pairs, respectively.

2) *ii) SCD Metrics:* The SeK coefficient and semantic change Kappa coefficient are adopted to evaluate the SCD performance, as in [9], [12], and [14]. The SeK coefficient, which is concerned with both the BCD and the semantic changes, is considered very close to the human evaluation of SCD results [9]. Meanwhile, as a part of the SeK metric, the semantic change Kappa coefficient is responsible for identifying the semantic changes. In this sense, if we suppose that the number of semantic change categories is N and the semantic change confusion matrix is $S \in \mathbb{R}^{N\times N}$, then the SeK coefficient and the semantic change Kappa are defined as follows:

$$\begin{aligned} \mathrm{SeK} &= e^{\mathrm{IoU}_{ch}-1}\cdot \mathrm{kappa} \\ \mathrm{kappa} &= \left(\widehat{\rho}-\widehat{\eta}\right)/\left(1-\widehat{\eta}\right) \end{aligned} \tag{20}$$

$$\widehat{\rho} = \left(\sum_{i=2}^{N} S_{ii}\right) \Big/ \left(\sum_{i=1}^{N}\sum_{j=1}^{N} S_{ij} - S_{11}\right) \tag{21}$$

$$\widehat{\eta} = \frac{\sum_{j=1}^{N} \widehat{S}_{j+}\cdot \widehat{S}_{+j}}{\left(\sum_{i=1}^{N}\sum_{j=1}^{N} S_{ij} - S_{11}\right)^2} \tag{22}$$

where S_{11} denotes the number of pixel pairs correctly predicted as unchanged. S_{j+} and S_{+j} denote the sum over the row and column of the confusion matrix S without S_{11}, respectively.

Three semantic consistency metrics based on IoU metric are introduced to evaluate SCD performance, to further help to understand the SCD methods [60]. Binary consistency (mIoU_{bc}) measures the consistency between the BCD result obtained by comparing the bitemporal LCC results ($\hat{Y}_1^{\mathrm{LCC}}$ and $\hat{Y}_2^{\mathrm{LCC}}$) and the BCD result ($\hat{Y}^{\mathrm{BCD}}$) directly output by the BCD decoder. Change area consistency (mIoU_{sc}) measures the consistency between the semantic change result and the ground truth in the changed areas

$$\mathrm{mIoU}_{sc} = \frac{1}{n_{\mathrm{change}}+1}\left(\sum_{i=0}^{n_{\mathrm{change}}} \mathrm{IoU}_i + \mathrm{IoU}_{uc}\right) \tag{23}$$

$$\mathrm{IoU}_{uc} = \mathrm{TN}/(\mathrm{TN}+\mathrm{FN}+\mathrm{FP}) \tag{24}$$

where n_{change} denotes the number of classes in the changed areas, and IoU_{uc} denotes the IoU of the unchanged area. No-change area consistency (mIoU_{nc}) measures the consistency between the semantic change result and the ground truth in the unchanged areas

$$\mathrm{mIoU}_{nc} = \frac{1}{n_{\text{no-change}}+1}\left(\sum_{i=0}^{n_{\text{no-change}}} \mathrm{IoU}_i + \mathrm{IoU}_{ch}\right) \tag{25}$$

where $n_{\text{no-change}}$ denotes the number of classes in the unchanged areas.

B. Experiment 1: The WHU Building Dataset

With a focus on building change, the WHU building change detection dataset covers Christchurch, New Zealand, where a 6.3-magnitude earthquake occurred in February 2011. The affected area was then rebuilt over the following years. The dataset contains a bitemporal remote sensing image pair of size 32 507 × 15 354, with a very high spatial resolution of 0.2 m. The dataset is annotated with buildings and nonbuildings, with four semantic change categories and one unchanged category to be detected. Along the side of 32 507 pixels in length and according to the ratio of 6:1:3, the dataset was divided into training, validation, and test sets, which were cropped into patches of size 512 × 512, without overlap. Finally, the WHU building dataset consisted of 1102, 174, and 551 bitemporal remote sensing image pairs for the training, validation, and test sets.

Fig. 6 shows several typical SCD visualization results achieved by the proposed SiamContrast on the WHU building dataset. As shown in Fig. 6(a1)–(b2), the big difference in radiation leads to false alarms for change in the SCD results of all the methods, but the SCD results of the proposed SiamContrast have fewer false alarms when compared with the other methods. Here, we note that there are many false alarms at the building edges in the results of SCDNet, because SCDNet uses the PCC mechanism to identify change and no change, which is sensitive to the land-cover edges, especially in bitemporal high-resolution images. As shown in Fig. 6(c1)–(d2), the SCD results achieved by the proposed SiamContrast have more regular edges and fewer incorrect predictions of semantic changes.

Table I lists the quantitative SCD results of the benchmark methods and the proposed SiamContrast on the WHU building

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Fig. 6. SCD visualization results on the WHU building dataset (■ buildings, ■ non-building, and *white:* non-change). Panels (a)–(d) represent four pairs of bi-temporal images, with "1" and "2" indicating the first and second time phases. The orange dashed boxes indicate locations of significant differences across various methods.

TABLE I
QUANTITATIVE RESULTS FOR THE WHU BUILDING DATASET

Methods	SCD metrics					BCD metrics	
	SeK	Kappa	mIoU_{bc}	mIoU_{sc}	mIoU_{nc}	F1	IoU_{ch}
HRSCD.str4	22.51	27.54	72.09	74.30	88.80	88.79	78.25
SCDNet	9.27	15.00	**100.00**	56.49	80.97	68.45	52.00
ChangeMask	28.70	33.73	74.70	77.46	**91.35**	**91.21**	**83.84**
SSCD-l	26.41	31.84	74.48	78.48	90.30	89.69	81.30
Bi-SRNet	28.88	34.02	76.51	**80.03**	91.00	91.09	83.63
SSESN	27.17	32.36	77.81	77.40	91.02	90.42	82.52
SFCCD	28.93	34.23	74.56	79.50	91.04	90.82	83.18
MCDnet	29.55	34.87	75.15	80.00	91.13	90.99	83.47
SCanNet	25.78	30.48	88.62	76.65	90.30	90.03	81.86
SiamContrast (proposed)	**29.78**	**35.14**	74.80	79.60	91.06	90.97	83.43

dataset. In terms of BCD metrics, the proposed SiamContrast method achieves competitive results compared to advanced methods. For SCD metrics, the proposed SiamContrast method attains the highest Kappa value, although it performs slightly lower than other methods in mIoU_{sc} and mIoU_{nc}. These results indicate that the proposed SiamContrast method excels in identifying major change categories but is less effective with minor change categories. Even so, SiamContrast outperforms other methods by 0.23%–20.51% in SeK, demonstrating the overall effectiveness of the proposed SiamContrast method.

Notably, SCDNet achieves a perfect score of 100% for binary consistency (mIoU_{bc}) because it directly compares bitemporal LCC results instead of using a BCD decoder to identify changes. Additionally, it can be observed that SCanNet achieves a lower Kappa value but significantly higher mIoU_{bc} than other methods. This discrepancy may

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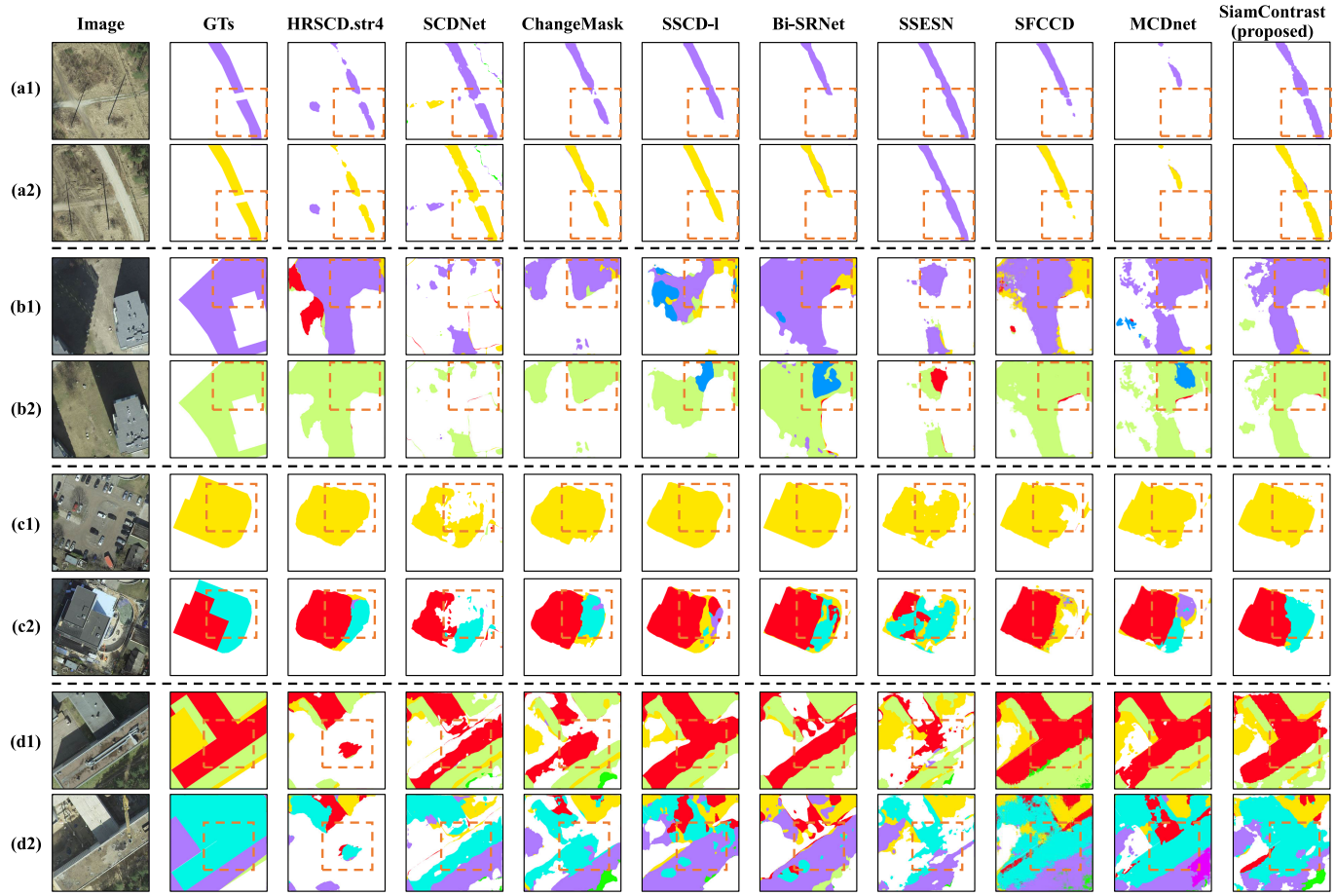


Fig. 7. SCD visualization results on the Hi-UCD mini dataset (■ water, ■ grass, ■ buildings, ■ greenhouse, ■ road, ■ bridge, ■ bare land, ■ woodland, ■ others, and *white:* non-change). Panels (a)–(d) represent four pairs of bi-temporal images, with "1" and "2" indicating the first and second time phases. The orange dashed boxes indicate locations of significant differences across various methods.

be attributed to the SCanFormer method used in SCanNet. SCanFormer is designed to consider spatiotemporal dependencies for semantic changes using BCD features and bitemporal LCC features. However, since the number of no-change pixels is much higher than that of change pixels, when land-cover labels in unchanged areas are also provided, SCanNet tends to capture the relationship between land-cover features and unchanged features. This may explain why the binary consistency of SCanNet is improved, while its SCD metrics remain limited.

C. Experiment 2: The Hi-UCD Mini Dataset

The Hi-UCD mini dataset [16], which belongs to Hi-UCD series,[1] is a well-annotated SCD dataset that covers a part of Tallinn, Estonia, with an area of 30 km^2 in 2017, 2018, and 2019. The Hi-UCD mini dataset focuses on nine urban land covers, i.e., water, grass, building, greenhouse, road, bridge, bare land, woodland, and others, which means potential 81 semantic change categories and one unchanged category to be detected. The Hi-UCD mini dataset contains 745 remote sensing image pairs of size 1024 × 1024, with a very high spatial resolution of 0.1 m. The bitemporal image pairs of the training set and the test set of the Hi-UCD mini dataset are from 2017 to 2018 and 2018 to 2019, respectively, with markedly different land-cover distributions. All the images were cropped into patches of size 512×512, without overlap, to obtain the Hi-UCD mini dataset. As a result, the Hi-UCD mini dataset consisted of 1200, 236, and 1544 bitemporal remote sensing image pairs for the training, validation, and test sets. Due to the many potential semantic change categories, the Hi-UCD mini dataset represents a challenge for the SCD task.

Fig. 7 exhibits the SCD visualization results of the proposed SiamContrast and the benchmark methods. As shown in Fig. 7(a1) and (a2), the similar colors of bare land and road make it difficult to locate change, while the results achieved by the proposed SiamContrast have fewer false alarms and false negatives of change when compared with those of the other methods. As shown in Fig. 7(b1) and (b2), the building shadows in the t_1 image result in the bare land being incorrectly classified as water, road, and building, leading to misclassification of the semantic change, but the proposed SiamContrast can significantly reduce these misclassifications. However, it should be noted that the results of SCDNet still show some false alarms for change at the land-cover edges, due to the PCC mechanism. Fig. 7(c1)–(d2) shows two complex change situations, where the proposed SiamContrast also shows its superiority in the identification of change and no change and the recognition of semantic changes.

[1]https://github.com/Daisy-7/Hi-UCD-S

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TABLE II
QUANTITATIVE RESULTS FOR THE Hi-UCD MINI DATASET

Methods	SCD metrics					BCD metrics		Params/M	FLOPs/G	Time cost/s	
	SeK	Kappa	$mIoU_{bc}$	$mIoU_{sc}$	$mIoU_{nc}$	F1	IoU_{ch}			Training	Test
HRSCD.str4	19.12	30.12	61.05	20.81	48.52	70.57	54.52	52.83	104.03	0.541	0.248
SCDNet	12.27	21.56	**100.00**	18.35	45.03	60.75	43.63	39.62	116.98	0.481	0.227
ChangeMask	22.39	34.06	62.55	21.29	43.48	73.45	58.04	33.93	94.43	0.527	0.242
SSCD-l	20.29	31.33	62.29	22.62	49.61	72.23	56.54	23.31	189.57	0.571	0.249
Bi-SRNet	18.23	28.40	64.78	22.86	45.46	71.53	55.68	23.39	189.91	0.630	0.267
SSESN	17.23	26.70	75.77	17.61	43.31	71.94	56.17	28.64	107.23	0.474	0.234
SFCCD	24.27	35.87	65.11	20.97	42.06	75.74	60.96	23.14	81.34	0.511	0.281
MCDnet	23.47	34.44	66.06	21.85	**51.05**	76.28	61.65	22.09	59.25	**0.401**	0.223
SCanNet	15.74	24.61	79.66	17.43	47.31	71.20	55.28	27.90	264.95	0.828	0.349
SiamContrast (proposed)	**25.75**	**37.67**	66.60	**25.31**	50.99	**76.51**	**61.96**	**9.12**	**57.81**	0.405 (0.059)	**0.199**

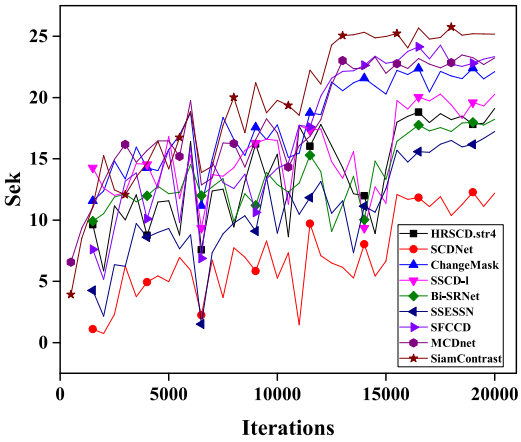


Fig. 8. SeK score curve of the proposed SiamContrast and the benchmark methods on the test set of the Hi-UCD mini dataset.

As shown in Table II, SiamContrast outperforms the other methods by 1.48%–13.48% in SeK, 1.80%–16.11% in Kappa, 2.5%–7.5% in $mIoU_{sc}$, 0.23%–15.76% in F1-score, and 0.23%–18.33% in IoU_{ch}. Notably, SSESN and SCanNet, both of which establish a relationship between change and land-cover features, achieve high scores in $mIoU_{bc}$ but low scores in $mIoU_{sc}$. These results are consistent with those observed in the WHU building dataset. In Table II, columns 9–10 show that the proposed SiamContrast has fewer parameters and requires less floating point operations (FLOPs). In Table II, columns 11–12 show the training and inference time costs for each batch across all methods. In the last row of Table II, it is evident that the proposed SiamContrast exhibits the shortest inference time, owing to its simple architecture. As for training time cost, the Siamese encoder and multitask decoders require 0.346 seconds, whereas SCD-CL necessitates 0.059 seconds. Even so, SiamContrast still maintains fast training speed. From Fig. 8, it can also be observed that the proposed SiamContrast also has a fast convergence speed, and its SeK curve is finally higher than the curves of all the other methods.

Moreover, t-SNE [64] is used to visualize the output of land-cover features by each method's decoder. As shown in Fig. 9, the features learned by the proposed SiamContrast are more compact than those of the other methods for each land-cover class, especially for roads and buildings. In this sense, the t-SNE results can account for the improvements of the proposed SiamContrast in the quantitative SCD performance metrics.

These experimental results further verify the superiority of SiamContrast, compared with the benchmark methods.

D. Experiment 3: The Hi-UCD Dataset

The Hi-UCD dataset [60], which is also part of the Hi-UCD series,[2] extends the Hi-UCD mini dataset. This dataset covers an area of 102 km^2 in Tallinn, Estonia, for the years 2018 and 2019. It shares the same nine urban land-cover categories as the Hi-UCD mini dataset and includes 48 types of semantic changes. The Hi-UCD dataset comprises 40,800 pairs of images, each with dimensions of 512 × 512 pixels and a spatial resolution of 0.1 m. In contrast, the Hi-UCD mini dataset consists of 12 000, 7200, and 21 600 bitemporal remote sensing image pairs for the training, validation, and test sets, respectively. The large data volume and detailed semantic annotations make the Hi-UCD dataset particularly challenging.

Fig. 10 presents the SCD visualization results of the proposed SiamContrast method alongside the benchmark methods. As illustrated in Fig. 10(a1)–(b2), the similar spectra of bare land, grass, and roads under shadow conditions make it challenging to detect changes and identify change categories. However, the results achieved by the proposed SiamContrast method can accurately identify most changes and correctly recognize the change categories compared to the other methods. As shown in Fig. 10(c1) and (c2), the presence of floating debris in the water bodies in the t_1 image results in the water being incorrectly classified as grass and bare land, leading to misclassification of the semantic change. However, the proposed SiamContrast method significantly reduces these misclassifications. Fig. 10(d1) and (d2) depict two complex

[2]https://github.com/Daisy-7/Hi-UCD-S

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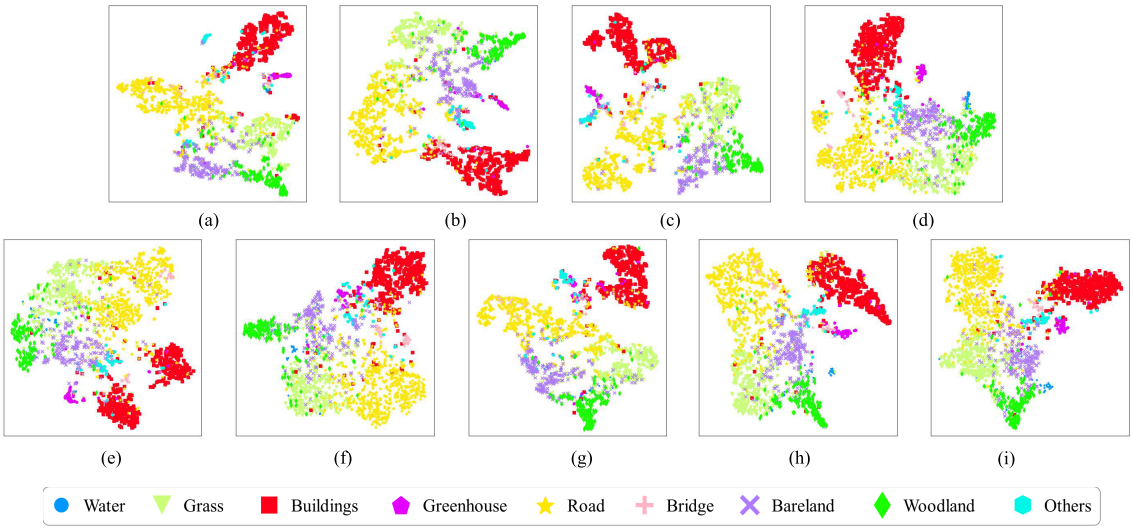


Fig. 9. T-SNE visualizations of the land-cover features output by the decoder of the proposed SiamContrast and the benchmark methods. (a) HRSCD.str4. (b) SCDNet. (c) ChangeMask. (d) SSCD-l. (e) Bi-SRNet. (f) SSESN. (g) SFCCD. (h) MCDNet. (i) SiamContrast (proposed).

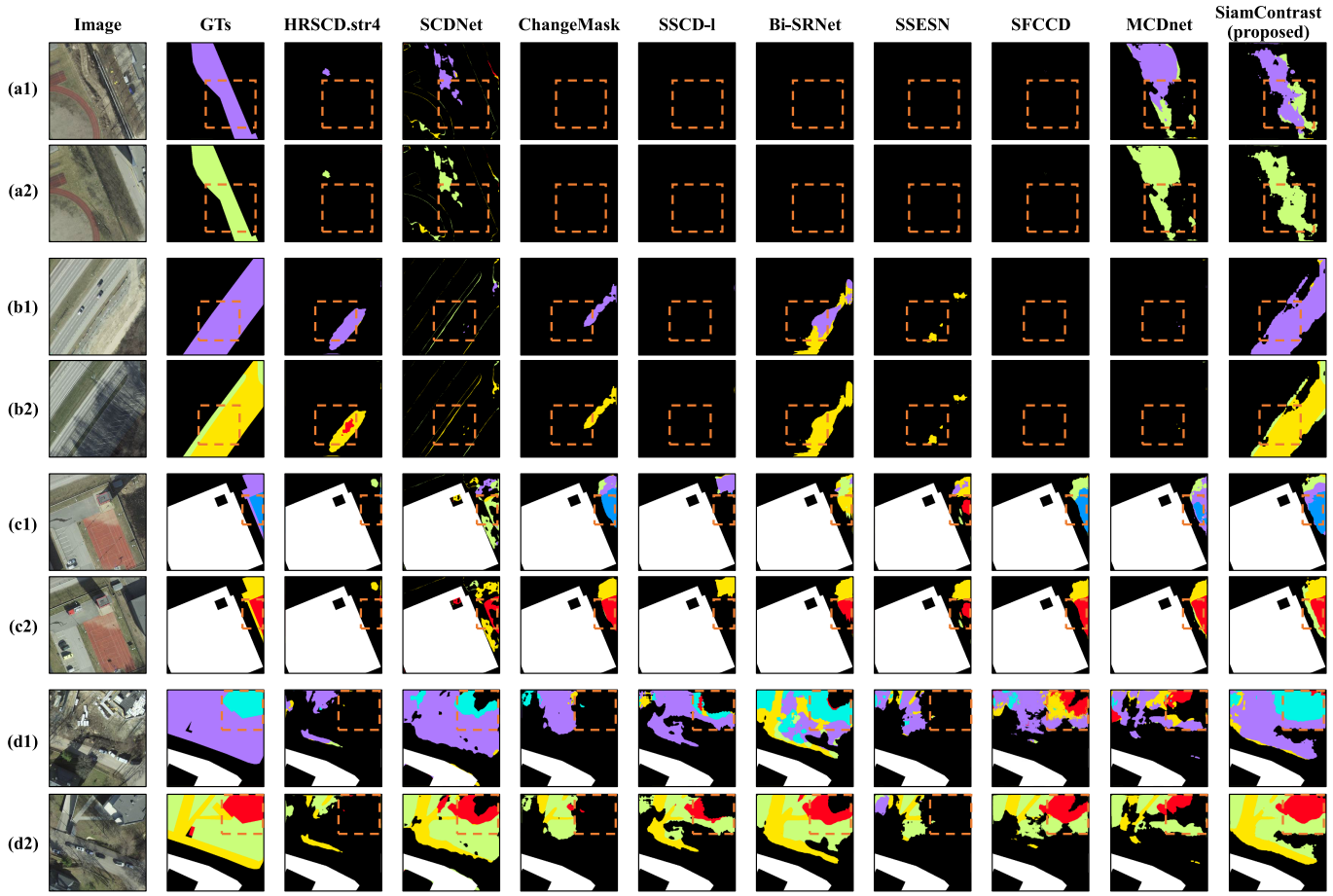


Fig. 10. SCD visualization results on the Hi-UCD dataset (■ non-change, ■ water, ■ grass, ■ buildings, ■ greenhouse, ■ road, ■ bridge, ■ bare land, ■ woodland, ■ others, and *white:* non-label). Panels (a)–(d) represent four pairs of bi-temporal images, with "1" and "2" indicating the first and second time phases. The orange dashed boxes indicate locations of significant differences across various methods.

change scenarios where the proposed SiamContrast method can correctly recognize semantic changes involving buildings and other objects with similar shapes.

As shown in Table III, SiamContrast outperforms the other methods by 2.96%–4.82% in SeK, 4.89–8.99% in Kappa, 1.08%–10.76% in mIoU_{sc}, 0.25%–7.65% in mIoU_{nc}, 1.25%–30.50% in F1-score, and 1.23%–23.79% in IoU_{ch}. Similar to the experimental results on the WHU building and Hi-UCD mini datasets, although SCanNet obtains competitive F1 and IoU_{ch} scores of BCD metrics, both SCanNet

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TABLE III
QUANTITATIVE RESULTS FOR THE Hi-UCD DATASET

Methods	SCD metrics					BCD metrics	
	SeK	Kappa	mIoU_{bc}	mIoU_{sc}	mIoU_{nc}	F1	IoU_{ch}
HRSCD.str4	2.75	5.79	45.80	14.13	52.38	40.75	25.60
SCDNet	-2.21	-5.25	**100.00**	7.48	49.37	23.68	13.43
ChangeMask	2.46	5.16	46.70	14.37	52.44	41.35	26.13
SSCD-l	3.58	7.36	45.59	12.13	47.87	43.57	27.88
Bi-SRNet	3.57	6.99	52.23	12.56	55.27	49.17	32.60
SSESN	-2.24	-4.55	60.67	7.76	50.10	45.02	29.05
SFCCD	2.19	4.29	47.74	17.16	48.60	49.21	32.66
MCDnet	2.45	4.77	50.78	16.35	54.76	49.88	33.23
SCanNet	1.72	3.26	61.60	8.70	54.84	52.93	35.99
SiamContrast (proposed)	**6.54**	**12.25**	49.89	**18.24**	**55.52**	**54.18**	**37.22**

and SSESN perform SCD involving the relationship between change and land-cover features, achieve high scores in mIoU_{bc} but low scores in mIoU_{sc} and Kappa. Notably, SCDNet and SSESN yield negative Kappa coefficients. This outcome is attributed to the relatively low prediction performance for change types and the highly imbalanced distribution of change types, which in turn results in negative SeK scores. A negative Kappa coefficient indicates significant inconsistency between the model's predicted change types and the actual change type labels, performing even worse than random guessing. In detail, the mIoU_{sc} values for SCDNet and SSESN are significantly lower than those of other methods, implying that the observed agreement $\hat{\rho}$ in the Kappa calculation [see (20)] is notably low. Due to the extreme imbalance in change type distribution, some change types have a disproportionately high number of samples, resulting in a higher expected agreement $\hat{\eta}$ in the Kappa calculation [see (20)]. Consequently, $\hat{\rho}-\hat{\eta}$ is less than 0, leading to a negative Kappa coefficient. These experimental results further demonstrate the effectiveness of the proposed SiamContrast.

V. ABLATION STUDY

In this section, we describe how we analyzed the different parts of the proposed method on the Hi-UCD mini dataset, to understand their influence when performing the SCD task.

A. Effect of the Different Parts

As shown in Table IV rows 2 and 3, the MCL in the proposed SCD-CL results in an improvement over the baseline Siamese network, with an SeK improvement of 1.74%, an F1-score improvement of 1.06%, and a Kappa improvement of 2.01%. It can be seen from Table IV rows 2–5 that only using the CCL can also bring improvements in the three scores, which are less than those of only using the MCL. Compared with the scores for the MCL, MCL+CCL brings slight improvements of 0.38% in the SeK score, 0.32% in the F1-score, and 0.41% in the Kappa score.

TABLE IV
EFFECT OF DIFFERENT PARTS OF SiamContrast ON THE Hi-UCD MINI DATASET

Method	SeK	F1	Kappa
Siam.	22.91	74.59	34.35
Siam.+MCL	24.65	75.65	36.46
Siam.+CCL	24.54	75.52	36.36
Siam.+MCL+CCL	25.03	75.97	36.87
Siam.+MCL+(CCL+SHAS)	24.15	75.39	35.85
SiamContrast (Siam.+MCL+(CCL+CHAS))	25.75	76.51	37.67

"Siam." denotes the proposed SiamContrast without any contrastive learning.

Rows 5 and 6 in Table IV demonstrate the effect of the existing SHAS strategy [27], which selects hard anchors from both changed and unchanged regions, when this strategy is incorporated into the CCL. However, the three scores in Table IV actually reduce. As shown in Table IV rows 5 and 7, when the proposed CHAS strategy, which emphasizes the importance of the hard anchors from changed regions, is applied to the CCL with the MCL, SeK, F1-score, and Kappa are improved by 0.72%, 0.54%, and 0.80%, respectively. These experiments validate the necessity of the CHAS strategy in disregarding hard anchors in unchanged regions, thereby preventing the network from excessively focusing on compressing intraclass distances across bitemporal images.

The ablation study verifies the effectiveness of the different parts of the proposed SCD-CL.

B. Decoder

In the DFP decoder, the absolute subtraction operator and FPN are responsible for modeling changes from bitemporal features and increasing the spatial resolution of BCD features, respectively. This section compares different approaches to modeling change and increasing feature resolution.

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TABLE V
COMPARISON BETWEEN DIFFERENT BASIC BCD FEATURE OPERATORS

Operator	SCD-CL	SCD metrics		BCD metrics	
		SeK	Kappa	F1	IoU_{ch}
temporally concatenation		19.58	31.10	69.89	53.72
	✓	21.28	33.30	71.14	55.21
temporal-symmetric transformer		23.40	34.91	75.02	60.03
	✓	23.92	35.28	75.89	61.15
absolute subtraction		22.91	34.35	74.59	59.49
	✓	**25.75**	**37.67**	**76.51**	**61.96**

TABLE VI
COMPARISON BETWEEN DECODER ARCHITECTURES

Decoder	SCD-CL	SCD metrics		BCD metrics	
		Sek	Kappa	F1	IoU_{ch}
FCN		24.17	35.94	75.26	60.34
	✓	24.51	36.42	75.31	60.39
UNet		22.63	34.00	74.44	59.29
	✓	23.44	35.09	74.72	59.65
FPN		22.91	34.35	74.59	59.49
	✓	**25.75**	**37.67**	**76.51**	**61.96**

1) Different BCD Operators: The absolute subtraction operator is a representative feature difference operator. To verify the effect of the feature difference operator, with the FPN fixed, three representative basic BCD feature operators were varied in the DFP decoder: temporally concatenation [29], temporal-symmetric transformer [12], and absolute subtraction [29]. The experimental results are shown in Table V. Rows 2, 4, and 6 of Table V show that the Siamese network with the temporally symmetric transformer achieves the best SCD and BCD performances than that of the other operators. However, when using the proposed SCD-CL to learn temporally invariant features, rows 3, 5, and 7 of Table V demonstrate that the Siamese network with the absolute subtraction obtains the best performances, improving the Siamese network baseline of the absolute subtraction by 2.84%, 3.32%, 1.92%, and 2.47% in SeK, Kappa, F1, and IoU_{ch} scores, respectively. It can also be found that the SCD-CL leads to greater improvements of the absolute subtraction in both the BCD and SCD scores compared to the temporal-symmetric transformer and temporally concatenated approaches when applied to the Siamese network. This can be attributed to the explicit modeling of the absolute subtraction between bitemporal features and BCD features, which aligns the goals of the BCD task, LCC task, and SCD-CL, i.e., reducing the distance of intraclass features and increasing the distance of interclass features.

2) Different Decoder Architectures: Two commonly used basic decoders for SCD, FCN [14], and U-Net [12], [15], [29] were compared with the FCN used in the SiamContrast. To verify the effect of different decoder architectures, while using the absolute subtraction to acquire BCD features, FCN, U-Net, and FPN were varied for both the BCD and LCC tasks.

As shown in Table VI rows 2, 4, and 6, when the SCD-CL is not used, the Siamese network with the FCN decoder achieves the best performances. This may be due to the match between the features of output stride 16 and the land covers in the Hi-UCD mini dataset. However, when the SCD-CL is used, Table VI shows that the improvements on U-Net and FPN are larger than that of FPN, which demonstrates that the influence of the SCD-CL is multiscale. Rows 4–7 of Table VI also show that the improvements from using FPN are larger than those of using U-Net. The difference in the approach of using multiscale features for prediction between U-Net and FPN may account for these different improvements. While U-Net progressively upsamples feature maps and uses the high resolution to predict, FPN upsamples multiscale feature maps to the same size to build a feature pyramid for prediction. This ensures that the multiscale features, improved by the SCD-CL, have an equal opportunity to contribute to the prediction.

The aforementioned experimental results verify that DFP and SiamFP decoders are suitable architectures for effectively utilizing the SCD-CL's ability in the SCD task.

C. Number of Projections for Each Land-Cover Category

M and M_c are the predefined numbers of selected projections for each land-cover category of a single-temporal image in the MCL and CCL, respectively. If M or M_c is too large, the hard anchors can be hard to account for a significant ratio for SCD-CL. In contrast, if M or M_c is too small, the diversity of the negative samples will not be guaranteed. In total, both too-large and too-small values of M or M_c can limit the effect of contrastive learning.

Because a typical value[3] of M in the literature [27] is 100 for semantic segmentation, the value of M was fixed to 100, and the value of M_c was varied over 50–200. As shown in Fig. 11(a), when M_c is 50, the SeK, F1-score, and Kappa reach the maximum simultaneously. In this context, the value of M_c was fixed to 50, and the value of M was varied over 50–200. Fig. 11(b) shows that the three scores reach the maximum values when M is 100. In summary, when $100 \leq M \leq 150$ in the MCL and $50 \leq M_c \leq 150$ in the CCL, the proposed SCD-CL can achieve a better SCD performance than the baseline Siamese network. As shown in Fig. 11(b), when $M = 50$ in the MCL, the three scores decrease to the extent that they are even worse than not using the MCL. The possible reason for this could be attributed to the limited positive and negative sample sampling, which introduces a strong bias toward anchors, resulting in a negative impact on the discrimination of the bitemporal land-cover feature representations. The score reduction does not happen to the CCL when $M_c = 50$, because the CCL uses bitemporal images while the MCL uses only single-temporal images. In this study, $M = 100$ and $M_c = 50$ were used in SiamContrast on both the WHU building and Hi-UCD mini datasets.

D. Hyperparameters for the Loss Balance

In the proposed SiamContrast, hyperparameter λ is used to balance SCD-CL and the cross-entropy losses of BCD and

[3]https://github.com/tfzhou/ContrastiveSeg/blob/main/configs/cityscapes/R_101_D_8.json

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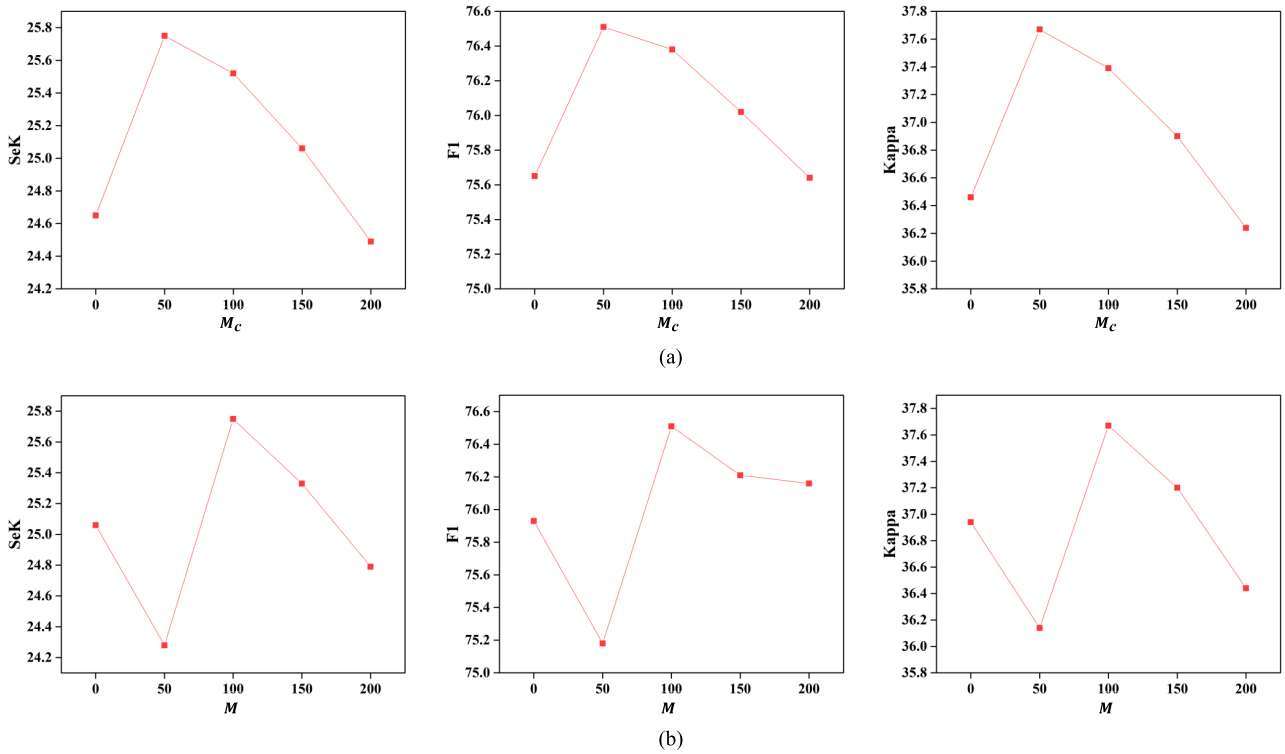


Fig. 11. Ablation study on M_c and M, which are the predefined numbers of selected projections for each land-cover category of a single-temporal image in the CCL and MCL, respectively. (a) SeK, F1-score, and Kappa with different values of M_c in the CCL. (b) SeK, F1-score, and Kappa with different values of M in the MCL.

bitemporal LCC, while hyperparameter μ is used to balance the MCL and CCL in SCD-CL.

As shown in Fig. 12(a), the SeK score reaches the maximum value when the balance coefficient λ of the proposed SCD-CL is set to the value of 0.1, which is also the typical value[3] in the literature [27]. As shown in Fig. 12(b), when μ is set to 1.0, which means that the MCL and CCL are equally weighted in SCD-CL, the SeK score is the highest.

VI. Limitation

The proposed SiamContrast was also evaluated using the widely used SECOND dataset [9], employing the same experimental settings as those used for the Hi-UCD mini and WHU building datasets. As shown in Table VII, both the full SiamContrast model and the baseline SiamContrast method without SCD-CL did not achieve strong SCD performance. Notably, the scores of the full SiamContrast model were even lower than those of the baseline. According to the analysis of Table V in Section V-B1, this is because the SiamContrast architecture is specifically designed for the discriminative features learned by SCD-CL. However, SCD-CL is ineffective on the SECOND dataset, which only provides annotated land-cover labels in changed regions. Due to the sparsity of change regions, there is a limited array of land-cover features. Consequently, the number of changed regions varies among different bitemporal image pairs, resulting in fluctuations in the number of available land-cover feature vectors. This variability leads to instability within both the MCL and CCL components of the SCD-CL method during training. Therefore, the utility of the proposed SCD-CL is limited when applied to SCD training sets that have land-cover annotations only within the changed regions. For SCanNet, owing to its utilization of SCanFormer and pseudo-label generation for unchanged areas, it achieves the best SCD performance on the commonly used SECOND dataset.

TABLE VII
Quantitative Results for the SECOND Dataset

Methods	SCD metrics				BCD metrics	
	SeK	Kappa	mIoU_{bc}	mIoU_{sc}	F1	IoU_{ch}
HRSCD.str4	18.79	29.99	50.99	24.94	69.47	53.23
SCDNet	6.39	13.62	**100.00**	13.42	39.19	24.37
ChangeMask	20.63	32.42	51.29	27.32	70.78	54.78
SSCD-l	21.50	33.09	58.23	**29.75**	72.52	56.88
SSESN	19.92	31.66	26.49	23.73	69.86	53.68
SFCCD	19.40	30.78	53.21	25.19	70.01	53.86
MCDnet	20.44	31.95	52.29	26.52	71.23	55.31
SCanNet	**23.94**	**36.21**	65.14	28.82	**73.89**	**58.60**
SiamContrast w.o. SCD-CL (proposed)	20.19	31.65	53.70	28.43	71.03	55.08
SiamContrast (proposed)	19.96	31.39	53.40	28.28	70.75	54.74

Recently, many researchers have manually annotated bitemporal remote sensing images [65] or modified the existing mapping products [16], [60], [66] to create SCD datasets with annotations for both changed and unchanged areas. This approach allows for the extensive use of a large number of

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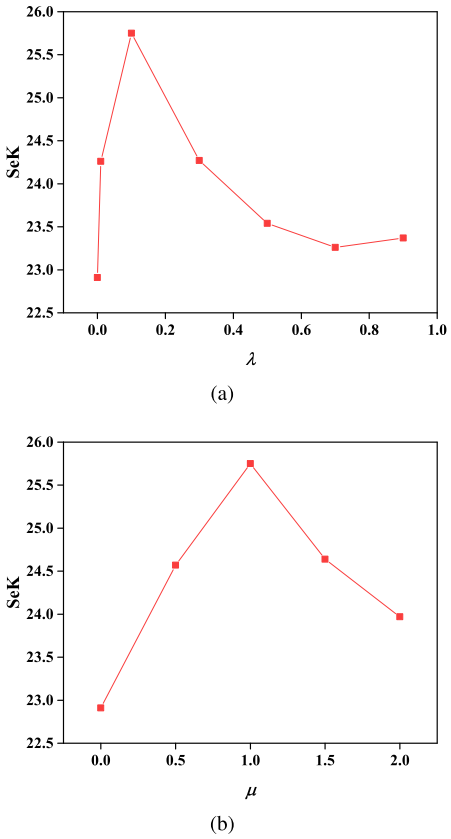


Fig. 12. SeK scores obtained for different predefined values of λ and μ in panels (a) and (b). Here, λ is used to balance SCD-CL and the cross-entropy losses of the BCD and bitemporal LCC, while μ is used to balance the MCL and CCL in SCD-CL.

land-cover samples in unchanged regions to enrich category annotations during training, which has great potential to help train high-performance SCD models. Consequently, designing SCD methods, such as the proposed SiamContrast method, specifically for SCD datasets that provide land-cover labels for both changed and unchanged areas is of significant importance.

VII. Conclusion

In this article, to overcome the land-cover feature cluster overlap problem caused by the large reflectance variability, the SiamContrast method is proposed for the SCD task. SiamContrast incorporates a novel SCD-CL method based on supervised contrastive learning to learn temporally invariant discriminative features. SCD-CL approach comprises two complementary components: MCL and CCL. MCL contrasts the features extracted from single-temporal images, enhancing the discrimination capability of the mono-temporal features. CCL contrasts the features across bitemporal images to align the features of the same class in two different time phases. Furthermore, the CHAS strategy specifically focuses the CCL part on hard samples from the changed regions, thereby improving the recognition of change types. The DFP decoder of the SiamContrast method employs feature distance modeling for binary change prediction, which fully exploits the discriminative abilities of bitemporal features. The extensive experimental results confirmed that the proposed SiamContrast can effectively improve the SCD performance, compared with the other advanced SCD methods. In the future, to obtain more negative samples, the memory bank technique [27], [44] will be used for SCD, a challenge of which may be designing a memory mechanism for different temporal features.

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